OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV-Semester (MAIN) Examinations July-2025

PRINCIPLES OF PROGRAMMING LANGUAGES

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What is the difference between Synthesized and Inherited Attributes?	2	1	1
b)	How does type conversion occur in mixed-mode assignments?	2	2	2
c)	What is the main advantage of using stack-dynamic local variables?	2	1	3
d)	Give a brief note about constructors in Java.	2	1	4
e)	List out the applications of logic programming.	2	1	5
f)	Distinguish Compiled and Interpreted Programming Languages?	2	4	1
2. a)	Discuss Language evaluation criteria.	4	2	1
b)	Explain the role of formal methods in describing the syntax of programming languages. How do context-free grammars help in defining the structure of programming constructs? Illustrate with suitable examples.	8	5	1
3. a)	How do different programming languages define and enforce the use of named constants? Explain with examples from at least two languages. Also, discuss any limitations or challenges associated with named constants.	6	2	2
b)	What are user-defined ordinal types in programming languages? Discuss their characteristics and uses, especially in languages like Pascal or Ada. How do they contribute to type safety and program clarity? Provide examples.	6	3	2

4. a)	Explain how nested selection statements are handled in modern programming languages. What are the issues related to readability and ambiguity (e.g., the dangling else problem), and how can these be resolved through language design or coding practices?	6	2	3	
b)	Explain the concept of a local referencing environment in subprograms. How are local variables, parameters, and non-local variables managed within a subprogram's scope? Discuss static and dynamic scoping in this context with suitable examples.	6	3	3	
5. a)	Explain the object-oriented extensions introduced in Ada 95. How does Ada implement type extension and polymorphism while maintaining its strong typing and safety goals?	6	4	4	
b)	Explain how exception handling is implemented in Ada. Discuss the exception block, predefined exceptions, and user-defined exceptions. How does Ada ensure program reliability and maintainability through its exception handling model?	6	2	4	
6. a)	Describe the fundamental features of functional programming languages, such as lambda calculus, recursion, lazy evaluation, and pattern matching. Explain how these features are used to build expressive and concise programs.	6	3	5	
b)	What is predicate calculus, and how is it used in logic programming languages? Explain the significance of predicates, quantifiers, and logical inference rules. How does predicate calculus serve as a foundation for languages like Prolog?	6	4	5	
7. a)	Describe the different types of type conversions in programming languages: implicit (coercion) and explicit (casting). How do type conversions affect expression evaluation and assignment statements? Discuss the potential risks involved in automatic type conversions, such as data loss or unexpected behavior.	6	2	2	2
b)	Discuss naming conventions and scope rules in scripting languages. What is the difference between global, local, and block scopes? Explain how closures and lexical scoping work in languages like JavaScript and Python. What challenges arise from these scoping mechanisms?	6	3	5	3